

CLINICAL TRIALS AND OBSERVATIONS

Genetics and MRD for therapy allocation in adults with Philadelphia chromosome–negative acute lymphoblastic leukemia

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KEY POINTS

- Genetics and MRD allow more accurate treatment with allo-HSCT or chemotherapy of adults with Ph[−] ALL.

In adults with Philadelphia chromosome–negative (Ph[−]) acute lymphoblastic leukemia (ALL) genetic risk is usually combined with measurable residual disease (MRD) assignment to consolidation therapy. However, this combination is not uniform across trials and centralized assessment is not always performed. This study analyzed patient outcomes using centrally assessed MRD and genetics. Patients with high genetic risk (HGR), those who required 2 induction cycles for complete remission (CR), and patients with CR with end-of-induction (EOI) MRD of $\geq 0.01\%$ were assigned to allogeneic hematopoietic stem cell transplant (allo-HSCT), whereas the remaining patients were assigned to delayed consolidation and maintenance.

HGR for B-cell ALL included *KMT2A* rearrangements, low hypodiploidy and age of >35 years, homozygous *TP53* mutations/deletions, or concomitant *IKZF1* and *CDKN2A/B* deletions. HGR in T-cell ALL (T-ALL) included absence of *NOTCH1/FBXW7* mutations and/or *K/NRAS* or *PTEN* alterations. Patients with early T-cell precursor (ETP) ALL received a different induction regimen, and all were assigned to allo-HSCT. Median age of 436 patients was 39 years (range, 18-60), 332 with B-lineage ALL and 104 with T-ALL. By intention to treat, 243 patients without ETP ALL (61%) were assigned to allo-HSCT and 157 (39%) to chemotherapy. The 3-year overall survival (OS) probability was 64% (95% confidence interval [CI], 58-69). For patients with CR and EOI MRD of <0.01% without HGR (n = 109), the OS probability was 81% (95% CI, 70-89), compared with 50% (95% CI, 34-63) for MRD-negative patients with HGR (n = 64). In patients with ETP ALL, the probability of 3-year OS was 61% (95% CI, 37-79). The combination of genetics and MRD allows accurate identification of adult patients with Ph⁻ ALL who are candidates for allo-HSCT or chemotherapy. This trial was registered at www.clinicaltrials.gov as NCT04179929.

Introduction

The prognosis of adult patients with acute lymphoblastic leukemia (ALL) depends on patient- and disease-related factors, although response to therapy remains the key determinant of relapse. Large risk-oriented adult ALL trials report survival rates of 50% to 70%. Measurable residual disease (MRD) is the strongest predictor of outcome,¹ with consistent prognostic value across ALL subtypes, time points, and detection methods.² Early achievement of MRD-negative complete remission (CR) at end-of-induction (EOI) has prognostic relevance in patients with Philadelphia chromosome-negative (Ph⁻) ALL,³ and most trials use MRD at this moment to assign patients to consolidation chemotherapy or allogeneic hematopoietic stem cell transplantation (allo-HSCT).⁴ However, MRD interpretation is influenced by ALL genetics,⁵ raising uncertainty about whether combining genetic risk and MRD improves identification of patients with Ph⁻ ALL who may benefit most from chemotherapy or allo-HSCT,^{6,7} a question made increasingly relevant by introduction of the bispecific monoclonal antibody blinatumomab into first-line Ph⁻ ALL therapy.⁸

The integration of MRD and genetics has been used for post-remission therapy assignment in several clinical studies. However, the genetic risk definition, the methods of genetic evaluation, and centralized vs local assessment are not uniform among trials. Furthermore, the combination with concurrent and centralized MRD measurement has not been performed in all studies. In 2019, the Spanish PETHEMA (Programa Español de Tratamientos en Hematología) group launched a trial in adults with Ph⁻ ALL, among whom postinduction therapy (consolidation or allo-HSCT) was assigned based on EOI MRD levels and patient genetic characteristics, both centrally assessed. Results of this study are reported herein.

Methods

Eligibility

Patients aged 18 to 60 years with de novo B-cell precursor (BCP) Ph⁻ ALL and T-cell ALL (T-ALL) were consecutively included in the PETHEMA ALL-2019 study across 94 Spanish hospitals. Exclusion criteria included performance status of >2 unrelated to ALL, *BCR::ABL1*-positive ALL, and mature B-cell ALL. The study began in December 2019, closed enrollment for this analysis in August 2024, and data was analyzed in April 2025. The protocol was approved by the Hospital Germans

Trias i Pujol ethics committee (reference PI-19-165) and was conducted according to the Declaration of Helsinki (Fortaleza, 2013). All patients provided written informed consent, and the trial was registered at www.clinicaltrials.gov as NCT 04179929.

Diagnostic procedures and definition of genetic risk

ALL was defined by >20% lymphoblasts in bone marrow (BM), together with immunophenotypic study performed by flow cytometry (FCM) using monoclonal antibodies reactive with B-cell, T-cell, myeloid, and precursor cell antigens. Early T-cell precursor (ETP) ALL was diagnosed by FCM, as previously reported.⁹ Conventional cytogenetics was performed locally and fluorescence in situ hybridization for *TCF3::PBX1* and *KMT2A* rearrangements and *TP53* deletions, was centralized. Centralized genomic studies included single-nucleotide polymorphism array (Cytoscan 750K, Thermo Fisher Scientific, Waltham, MA) and next-generation sequencing using customized panels for BCP-ALL (Illumina, San Diego, CA) and T-ALL (Agilent, Santa Clara, CA) developed at Hospital Universitario de Salamanca (Salamanca, Spain) and the Josep Carreras Research Institute (Badalona, Spain), respectively. *BCR::ABL1*-like ALL was identified using a commercial fluorescence in situ hybridization panel (Milton, Cambridge, United Kingdom) and a customized targeted RNA-sequencing panel (Ion Torrent, Guilford, CO) developed at Hospital 12 de Octubre (Madrid, Spain).¹⁰

High genetic risk (HGR) for BCP-ALL was defined as the presence of any of the following: *KMT2A* rearrangements, low hypodiploidy (30-39 chromosomes) and age of >35 years, biallelic *TP53* alterations, or concomitant *IKZF1* and *CDKN2A/B* deletions.¹¹⁻¹³ Patients with low-hypodiploidy and aged <35 years were not considered as having HGR despite having *TP53* biallelic alteration.¹³ In this trial, patients with *BCR::ABL1*-like were not considered as having HGR. HGR for T-ALL included the absence of *NOTCH1/FBXW7* mutations and/or the presence of *K/NRAS* or *PTEN* alterations.¹⁴

MRD assessment

MRD levels in the BM were centrally assessed at the following time points: day 14 of the first induction cycle (I-1), at EOI-1 (weeks 5-6) or EOI-2 (I-2) if necessary, at the end of the third cycle of early consolidation and reinduction (weeks 20-22), and before allo-HSCT. The EuroFlow operating procedures and 2-tube, 12-color FCM panels for BCP-ALL and T-ALL, respectively, were used.¹⁵ For data analysis the Infinicyt software (BD-Cytognos, SL,

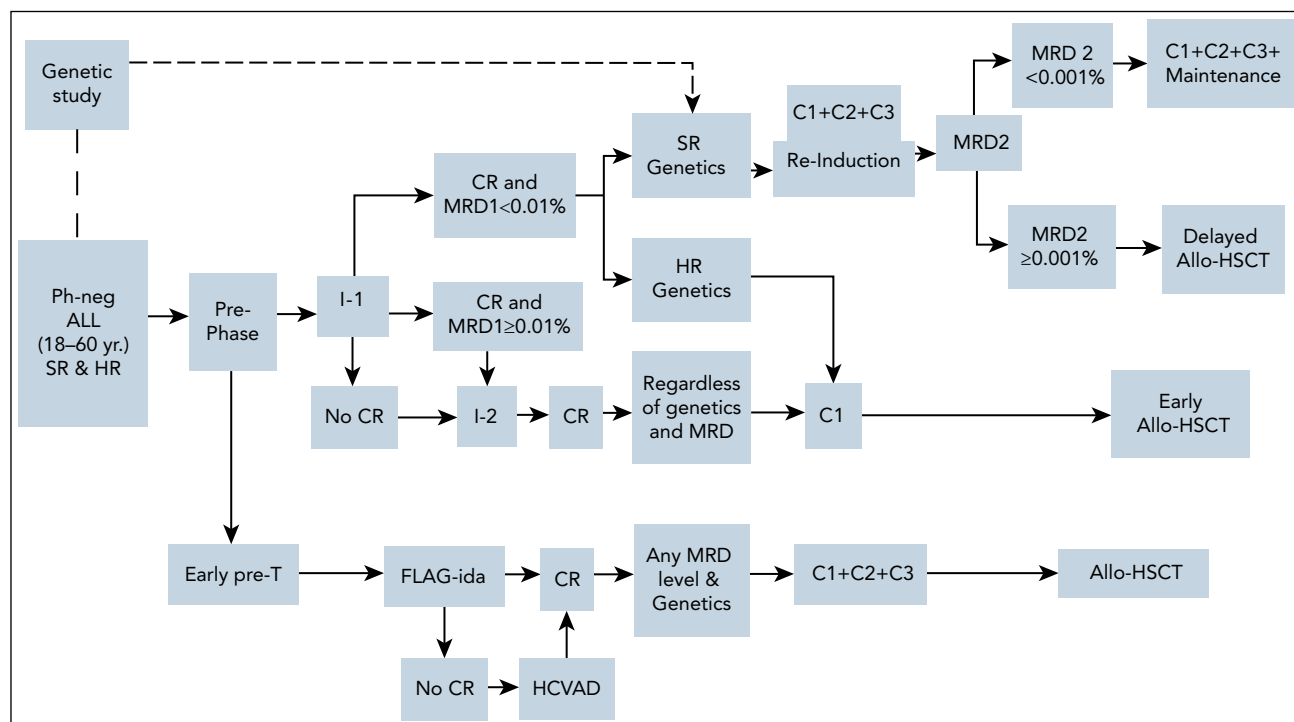


Figure 1. Schedule of the PETHEMA LAL-2019 trial. Patients with complete response after I-1 who achieved MRD levels of <0.01% and had SGR received early consolidation and reinduction. If MRD levels were <0.001% they continued with delayed consolidation and maintenance. The remaining patients were allocated to early or delayed allo-HSCT. Patients with ETP-ALL received a differentiated induction therapy and were assigned to receive allo-HSCT. C1/2/3, cycle 1/2/3; HR, high risk; SR, standard risk.

Salamanca, Spain) was used.^{16,17} The limit of detection and the limit of quantitation of the method were of 2×10^{-6} and 5×10^{-6} , respectively.

Treatment

Figure 1 outlines the trial schedule, and Table 1 summarizes the chemotherapy regimen. I-1 included vincristine, prednisone, daunorubicin, and pegylated *Escherichia coli* asparaginase (PEG-ASP; 1500 IU/m²). Patients without CR or with EOI-1 MRD of $\geq 0.01\%$ received FLAG-Ida (fludarabine, cytarabine [ARA-C], granulocyte colony-stimulating factor, and idarubicin). A June 2022 amendment allowed inotuzumab ozogamicin (InO) to replace FLAG-Ida in patients with refractory BCP-ALL after I-1. Patients who did not achieve CR after I-1 and I-2 were excluded. Patients in CR without HGR and MRD of <0.01% received early consolidation with 3 cycles of high-dose methotrexate; high-dose ARA-C and PEG-ASP; followed by 1 reinduction cycle with vincristine, prednisone, and PEG-ASP. If MRD was <0.001% after those 3 cycles, delayed consolidation followed by maintenance therapy was administered for up to 2 years in CR.

Patients with ETP-ALL received FLAG-Ida as I-1 (Table 1). HyperCVAD (hyperfractionated cyclophosphamide, vincristine, doxorubicin, and dexamethasone), alternating with high-dose methotrexate and ARA-C, was administered as I-2 to patients with resistant disease (RD). Patients in CR after I-1 or I-2 received the same aforementioned 3 early consolidation cycles, followed by allo-HSCT, regardless of MRD levels or genetic risk. Patients without CR after I-1 and I-2 were excluded.

Central nervous system (CNS) prophylaxis included 14 triple intrathecal chemotherapy (TIT) doses, also recommended before and after allo-HSCT. Patients with CNS involvement at diagnosis received TIT every 3 to 4 days until cerebrospinal fluid clearance in 2 consecutive samples, with a minimum of 5 TIT administrations. Then, patients followed the regular CNS prophylaxis. No CNS radiotherapy was administered.

For patients aged >50 years, doses were reduced to methotrexate, 1.5 g/m²; ARA-C, 1000 mg/m² every 12 hours; and PEG-ASP, 500 IU/m² (Table 1). PEG-ASP 500 IU/m² was given to patients with body mass index of >30, hepatic steatosis or fibrosis, or previous liver disease. ASP activity was centrally assessed; levels of >0.1 IU/mL at 2 weeks were considered therapeutic. *Erwinia* ASP replaced PEG-ASP in cases of hypersensitivity or silent inactivation. Toxicities during induction and early consolidation/reinduction were graded according to the common terminology criteria for adverse events, version 5.0.

Risk-oriented therapy

Criteria for assignment to allo-HSCT were any of the following: (1) HGR regardless of MRD level; (2) requirement of 2 induction cycles to achieve CR; (3) CR and MRD of $\geq 0.01\%$ at EOI-1; (4) MRD of $\geq 0.001\%$ at the end of early consolidation/reinduction; and (5) any patient with ETP-ALL. The remaining patients were assigned to consolidation and maintenance chemotherapy. For allo-HSCT, all stem cell sources and conditioning regimens were permitted according to national and international guidelines, patient characteristics, and the responsible

Table 1. PETHEMA ALL-2019 trial: chemotherapy schedule

Phase	Route	Dose	Days
BCP-ALL and non-ETP ALL			
I-1*			
Vincristine (maximum 2 mg)	IV	1.5 mg/m ²	1, 8, 15, 22
Daunorubicin	IV	45 mg/m ²	1-3
Prednisone	IV	60 mg/m ²	1-14
	IV	30 mg/m ²	15-21
	IV	15 mg/m ²	22-28
PEG-ASP†,‡	IV	1500 IU/m ²	15, 29
TIT§	IT		1, 22
I-2 (FLAG-Ida)			
Idarubicin	IV	12 mg/m ²	1, 3, 5
Fludarabine	IV	30 mg/m ²	1-5
ARA-C	IV	2000 mg/m ²	1-5
TIT§	IT		7
I-2 (inotuzumab)¶			
Inotuzumab	IV	0.8 mg/m ² ; 0.5 mg/m ² ; 0.5 mg/m ²	1, 8, 15
Inotuzumab#	IV	0.5 mg/m ²	1, 8, 15
Early consolidation-1			
Vincristine	IV	2 mg	1, 8
Dexamethasone	IV/PO	20 mg/m ²	1-5
	IV/PO	10 mg/m ²	6
	IV/PO	5 mg/m ²	7
	IV/PO	2.5 mg/m ²	8
Methotrexate**	IV	3 or 5 g/m ²	1
PEG-ASP†,‡	IV	1500 IU/m ²	3
TIT§	IT		1
Early consolidation-2			
Dexamethasone	IV/PO	20 mg/m ²	1-5
	IV/PO	10 mg/m ²	6
	IV/PO	5 mg/m ²	7
	IV/PO	2.5 mg/m ²	8
ARA-C	IV	2000 mg/m ² every 12 h	1, 2
PEG-ASP†,‡	IV	1500 IU/m ²	3
TIT§	IT		4
Early consolidation-3			
Vincristine	IV	2 mg	1, 8
Dexamethasone	IV/PO	20 mg/m ²	1-5

BMI, body mass index; IM, intramuscular; IT, intrathecal; PO, per os (orally).

*Prephase with prednisone 60 mg/m² and TIT was given for a maximum of 1 week, while ALL was fully characterized.

†Dose reduction to 500 IU/m² in patients aged >50 years, with BMI of >30, liver steatosis or fibrosis, or preexisting liver disease.

‡Change to Erwinia ASP in cases of hypersensitivity or silent inactivation. Each dose of PEG-ASP will be substituted by 25 000 IU/m², 3 doses per week, for 2 weeks (total: 6 doses); 50% dose reduction in patients aged >50 years.

§TIT with methotrexate (15 mg), ARA-C (30 mg), and hydrocortisone (20 mg).

||50% dose reduction in patients aged >50 years.

¶After amendment in June 2022.

#The second cycle is only for patients in CR but MRD level of ≥0.05%.

**3 g/m² for B-cell-precursor ALL, 5 g/m² for T-ALL. Dose reduction to 1.5 g/m² for patients aged >50 years.

††Only for patients assigned to receive chemotherapy.

‡‡Until completing 1 year from the date of the CR.

^aEvery month.

^bDuring the second year from the date of the CR.

^cOnly if no CR or MRD of ≥0.01% after FLAG-Ida; 1 cycle only.

Table 1 (continued)

Phase	Route	Dose	Days
	IV/PO	10 mg/m ²	6
	IV/PO	5 mg/m ²	7
	IV/PO	2.5 mg/m ²	8
Methotrexate**	IV	3 or 5 g/m ²	1
PEG-ASP†,‡	IV	1500 IU/m ²	3
TIT§	IT		1
Reinduction			
Vincristine (maximum 2 mg)	IV	1.5 mg/m ²	1, 8, 15, 22
Prednisone	IV	60 mg/m ²	1-14
	IV	30 mg/m ²	15-21
PEG-ASP†,‡	IV	1500 IU/m ²	15
Delayed consolidation-1††			
Vincristine	IV	2 mg	1, 8
Dexamethasone	IV/PO	20 mg/m ²	1-5
	IV/PO	10 mg/m ²	6
	IV/PO	5 mg/m ²	7
	IV/PO	2.5 mg/m ²	8
Methotrexate**	IV	3 or 5 g/m ²	1
PEG-ASP†,‡	IV	1500 IU/m ²	3
TIT§	IT		1
Delayed consolidation-2††			
Dexamethasone	IV/PO	20 mg/m ²	1-5
	IV/PO	10 mg/m ²	6
	IV/PO	5 mg/m ²	7
	IV/PO	2.5 mg/m ²	8
ARA-C	IV	2000 mg/m ² every 12 h	1, 2
PEG-ASP†,‡	IV	1500 IU/m ²	3
TIT§	IT		4
Delayed consolidation-3††			
Vincristine	IV	2 mg	1, 8
Dexamethasone	IV/PO	20 mg/m ²	1-5
	IV/PO	10 mg/m ²	6
	IV/PO	5 mg/m ²	7
	IV/PO	2.5 mg/m ²	8
Methotrexate**	IV	3 or 5 g/m ²	1
PEG-ASP†,‡	IV	1500 IU/m ²	3
TIT§	IT		1

BMI, body mass index; IM, intramuscular; IT, intrathecal; PO, per os (orally).

*Prephase with prednisone 60 mg/m² and TIT was given for a maximum of 1 week, while ALL was fully characterized.

†Dose reduction to 500 IU/m² in patients aged >50 years, with BMI of >30, liver steatosis or fibrosis, or preexisting liver disease.

‡Change to *Erwinia* ASP in cases of hypersensitivity or silent inactivation. Each dose of PEG-ASP will be substituted by 25 000 IU/m², 3 doses per week, for 2 weeks (total: 6 doses); 50% dose reduction in patients aged >50 years.

§TIT with methotrexate (15 mg), ARA-C (30 mg), and hydrocortisone (20 mg).

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‡‡Until completing 1 year from the date of the CR.

§Every month.

¶During the second year from the date of the CR.

||Only if no CR or MRD of ≥0.01% after FLAG-Ida; 1 cycle only.

Table 1 (continued)

Phase	Route	Dose	Days
Maintenance-1 ^{†‡}			
Mercaptopurine	PO	50 mg/m ²	Daily
Methotrexate	IM	20 mg/m ²	Weekly
Reinductions ^a			
Vincristine	IV	2 mg	1
Prednisone	PO	60 mg	1-7
TIT [§]	IT		1
Maintenance-2 ^b			
Mercaptopurine	PO	50 mg/m ²	Daily
Methotrexate	IM	20 mg/m ²	Weekly
ETP-ALL			
I-1 (FLAG-Ida)			
Idarubicin	IV	12 mg/m ²	1, 3, 5
Fludarabine	IV	30 mg/m ²	1-5
ARA-C	IV	2000 mg/m ²	1-5
TIT [§]	IT		7
I-2 (HyperCVAD) ^c			
Part A			
Cyclophosphamide	IV	300 mg/m ²	1-3
Vincristine	IV	2 mg	4, 11
Doxorubicin	IV	50 mg/m ²	4
Dexamethasone	IV	40 mg	1-4, 11-14
TIT [§]	IT		7
Part B			
Methotrexate	IV	1000 mg/m ²	1
ARA-C	IV	3000 mg/m ² every 12 h	2, 3
Early consolidation-1			
Vincristine	IV	2 mg	1, 8
Dexamethasone	IV/PO	20 mg/m ²	1-5
	IV/PO	10 mg/m ²	6
	IV/PO	5 mg/m ²	7
	IV/PO	2.5 mg/m ²	8
Methotrexate ^{**}	IV	3 or 5 g/m ²	1
PEG-ASP ^{†,‡}	IV	1500 IU/m ²	3
TIT [§]	IT		1
Early consolidation-2			
Dexamethasone	IV/PO	20 mg/m ²	1-5

BMI, body mass index; IM, intramuscular; IT, intrathecal; PO, per os (orally).

*Prephase with prednisone 60 mg/m² and TIT was given for a maximum of 1 week, while ALL was fully characterized.

†Dose reduction to 500 IU/m² in patients aged >50 years, with BMI of >30, liver steatosis or fibrosis, or preexisting liver disease.

‡Change to Erwinia ASP in cases of hypersensitivity or silent inactivation. Each dose of PEG-ASP will be substituted by 25 000 IU/m², 3 doses per week, for 2 weeks (total: 6 doses); 50% dose reduction in patients aged >50 years.

§TIT with methotrexate (15 mg), ARA-C (30 mg), and hydrocortisone (20 mg).

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^aEvery month.

^bDuring the second year from the date of the CR.

^cOnly if no CR or MRD of ≥0.01% after FLAG-Ida; 1 cycle only.

Table 1 (continued)

Phase	Route	Dose	Days
ARA-CII PEG-ASP†,‡ TIT§ Early consolidation-3 Vincristine Dexamethasone Methotrexate** PEG-ASP†,‡ TIT§	IV/PO	10 mg/m ²	6
	IV/PO	5 mg/m ²	7
	IV/PO	2.5 mg/m ²	8
	IV	2000 mg/m ² every 12 h	1, 2
	IV	1500 IU/m ²	3
	IT		4
	IV	2 mg	1, 8
	IV/PO	20 mg/m ²	1-5
	IV/PO	10 mg/m ²	6
	IV/PO	5 mg/m ²	7
	IV/PO	2.5 mg/m ²	8
	IV	3 or 5 g/m ²	1
	IV	1500 IU/m ²	3
	IT		1

BMI, body mass index; IM, intramuscular; IT, intrathecal; PO, per os (orally).

*Prephase with prednisone 60 mg/m² and TIT was given for a maximum of 1 week, while ALL was fully characterized.

†Dose reduction to 500 IU/m² in patients aged >50 years, with BMI of >30, liver steatosis or fibrosis, or preexisting liver disease.

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¶After amendment in June 2022.

#The second cycle is only for patients in CR but MRD level of ≥0.05%.

**3 g/m² for B-cell-precursor ALL, 5 g/m² for T-ALL. Dose reduction to 1.5 g/m² for patients aged >50 years.

††Only for patients assigned to receive chemotherapy.

‡‡Until completing 1 year from the date of the CR.

*Every month.

^bDuring the second year from the date of the CR.

^cOnly if no CR or MRD of ≥0.01% after FLAG-Ida; 1 cycle only.

physician's decisions. In cases without available genetic risk assessment, assignment was based on MRD levels.

Outcome definitions

CR was defined as <5% BM blasts, no extramedullary disease, neutrophil count of >1 × 10⁹/L and platelet count of >100 × 10⁹/L. Poor MRD response was defined as MRD of ≥0.01% after I-1 or ≥0.001% after early consolidation/reinduction. RD was defined as persistence of leukemia in patients surviving induction, and early death (ED) was death before fulfilling the criteria for CR or RD. Relapse was defined as disease recurrence at any site after CR. Overall survival (OS) was measured from protocol entry to death or last follow-up. Event-free survival (EFS) was calculated from diagnosis to failure, relapse, or death by any cause, whichever occurred first. Relapse-free survival (RFS) was estimated from first CR to relapse or death by any cause, whichever occurred first. The cumulative incidence of relapse (CIR) was calculated from first CR to relapse, considering nonrelapse mortality as a competing event.

Study objectives and statistical analysis

The primary objective was OS based on the 2 postinduction treatment options evaluated by intention to treat (ITT).

Secondary objectives were rates of CR and MRD response, EFS, RFS, CIR assessed by ITT, and CIR and OS per protocol, excluding protocol deviations. Comparison with outcomes of similar patients included in the preceding ALL-AR-11 trial⁴ was also performed. The main clinical and hematological variables in the whole series, and in the different treatment subgroups, were compared by the median test (continuous variables) and Pearson or Fisher exact tests (categorical variables). OS curves were plotted by the Kaplan-Meier method and compared with the log-rank test. CIR was estimated using cumulative incidence functions via competing risks analysis. The Gray test was used for comparison of CIR curves. Multi-variable analyses for OS and CIR were performed using the Cox proportional hazards regression model and the Fine and Gray model, respectively. No imputation method was used for missing data. IBM SPSS (version 24) and R software (version 4.5.0) were used for statistical analyses. Two-sided *P* values of <.05 were considered statistically significant.

Results

Patient entry

From December 2019 to August 2024, 436 patients aged 18 to 60 years with Ph⁻ ALL, from 94 Spanish hospitals entered the study and were the focus of the analysis.

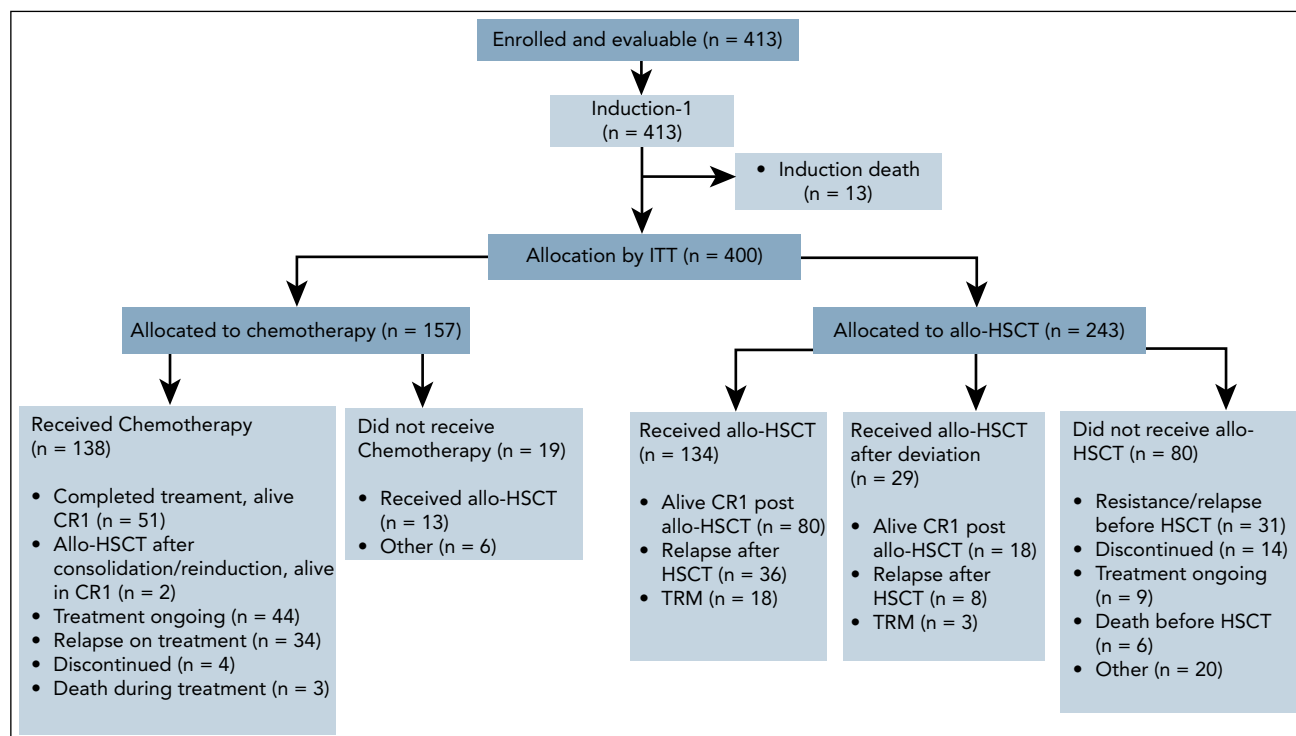


Figure 2. Flowchart of the PETHEMA ALL-2019 study (B-ALL and non-ETP ALL). Patients who survived the first induction therapy (I-1), achieved CR with MRD level of <0.01% and had SGR were allocated to the chemotherapy arm by ITT. Patients with disease refractory to I-1, those with CR after I-1 with MRD of $\geq 0.01\%$, or those with HGR independent of MRD levels were allocated to the allo-HSCT arm by ITT. CR1, first CR.

Pretreatment characteristics and genetic risk assignment

Figure 2 shows the study flowchart. The main demographic and clinical characteristics are summarized in Table 2. Of 436 patients, 246 (56%) were male. The median (range) age was 39 (18-60) years; 234 (54%) patients were aged 18 to 40 years. Extramedullary disease was observed in 105 of 402 (26%) patients. BCP-ALL was diagnosed in 332 (76%) patients, and T-ALL in 104 (24%; 23 of whom had ETP-ALL).

Table 3 shows the main genetic groups among 282 patients with BCP-ALL. *BCR::ABL1*-like (n = 41 [15%]), *KMT2Ar* (n = 30 [11%]), low hypodiploid (n = 22 [8%]), high hyperdiploid (n = 18 [6%]), *TCF3::PBX1* (n = 14 [5%]), and *PAX5 P80R* (n = 14 [5%]) were the most prevalent. Their main characteristics and outcomes are shown in Table 4. *IKZF1* deletions were observed in 130 of 266 cases (49%), *CDNK2A/B* deletions in 109 of 266 (41%), and homozygous *TP53* mutations/deletions in 20 of 266 (8%). The *IKZF1*plus signature¹⁸ was detected in 74 of 266 cases (28%). In T-ALL, *NOTCH1/FBXW7* mutations were found in 64 of 103 (62%) cases, *K/NRAS* alterations in 14 of 103 (14%) cases and *PTEN* alterations in 19 of 100 (19%).

Genetic risk could be assessed in 358 of 436 (82%) patients, 152 (43%) of whom showed HGR (99/274 [36%] for BCP-ALL and 53/84 [63%] for T-ALL; Table 4). All patients with ETP ALL showed HGR.

Response to induction

The main results after I-1 in patients who did not have ETP-ALL included: ED in 13 patients (3%), RD in 52 (13%), and CR in 348

(84%). No differences in CR rates were observed according to age (84.3% vs 84% for patients younger or older than 40 years, respectively) and immunophenotype (84.6% for BCP-ALL vs 83% for T-ALL). CR was less frequent in patients with pre-T-ALL compared with those with cortical and mature T-ALL (68% vs 95%, respectively; $P = .002$), whereas no differences were observed in BCP-ALL immunologic subtypes. MRD <0.01% after I-1 was attained in 221 of 348 (63%) patients and MRD <0.001% was reached in 149 of 333 (45%). I-2 was scheduled for 179 patients (52 with RD, and 127 in CR but MRD of $\geq 0.01\%$). FLAG-Ida was administered to 138 patients (105 in CR with MRD $\geq 0.01\%$, and 33/46 with disease refractory to I-1), whereas InO was administered in 13 of 46 patients with disease refractory to I-1. The remaining 28 patients did not receive the second induction therapy despite being indicated (n = 12) or were treated with regimens not prescribed by the protocol (n = 16). The death rate after FLAG-Ida was 4% (6/138), no deaths occurred in patients receiving InO, and the frequencies of CR attainment in patients with disease refractory to I-1 treated with FLAG-Ida or InO were of 82% (86% for BCP-ALL) and 92%, respectively, with a rate of MRD <0.01% of 54% vs 67%, respectively. At EO1-2, 103 patients were in CR with MRD of <0.01%. In summary, after I-1 and I-2, CR was achieved in 387 of 407 (95%) patients, and 324 of 357 (91%) showed MRD of <0.01%. Among patients with RD, 39 had BCP-ALL (13 B-other; 9 *BCR::ABL1*-like; 6 low hypodiploid; 2 *PAX5* alt and *ZNF384r MYC*, *CEBP/ZEB2*, and *IGHr*, 1 case each; and 5 patients not evaluable) and 13 T-ALL (12 pre-T-ALL, and 1 cortical).

The main results after FLAG-IDA among patients with ETP-ALL included: ED in 1 patient (4%), RD in 5 (22%), and CR in 17 (74%). MRD of <0.01% was achieved in 7 patients (41%).

Table 2. Clinical and biological characteristics of patients with ALL according to the phenotype and the post-I-1 therapeutic option

Characteristic	Whole series (N = 436)	BCP-ALL (n = 332)	T-ALL (n = 104)	Chemotherapy (n = 157)	Allo-HSCT (n = 243)
Age (y)					
Median (range)	39 (18-60)	40.5 (18-60)*	33 (18-60)*	37 (18-60)	40 (18-60)
18-40, n (%)	234 (54)	166 (50)†	68 (65)†	90 (57)	124 (51)
Sex, n (%)					
Male	246 (56)	178 (54)‡	68 (65)‡	91 (58)	134 (55)
Female	190 (44)	154 (46)‡	36 (35)‡	66 (42)	109 (45)
Hepatomegaly	65/391 (17)	47/293 (16)	18/98 (18)	21/137 (15)	39/223 (18)
Splenomegaly	115/397 (29)	78/298 (26)§	37/99 (37)§	40/141 (28)	68/225 (30)
Lymphadenopathy	102/397 (26)	55/301 (18)*	47/96 (49)*	44/143 (31)¶	48/224 (21)¶
Extramedullary involvement					
Mediastinum	41/399 (10)	3/300 (1)*	38/99 (38)*	17/142 (12)	19/228 (8)
CNS	52/402 (13)	35/305 (11)	17/97 (17)	11/143 (8)	33/229 (14)
Testicle	2/216 (1)	2/155 (1)	0	0	1/118 (1)
Other	26/402 (6)	12/304 (4)*	14/98 (14)*	9/141 (6)	15/232 (7)
WBC count					
Median (range), ×10 ⁹ /L	12.7 (0.4-740)	8.8 (0.4 -720)*	29 (1.2-740)*	8.9 (0.4-551.4)¶	14.8 (0.6-740)¶
>30 × 10 ⁹ /L	124/428 (29)	76/326 (23)*	48/102 (47)*	27/155 (17)*	86/239 (36)*
>100 × 10 ⁹ /L	67/428 (16)	44/326 (14)#	23/102 (22)#	6/155 (4)*	54/239 (23)*
Hemoglobin, median, range, g/dL	98 (23-178)	93 (23-160)*	115 (38-178)*	100 (23-178)	97 (28-171)
Platelets, median, range, g/dL	56.5 (2-680)	54 (3-680)	63 (2-575)	59 (4-575)	52 (2-680)
BM blast cells, median, range, %	88 (20-100)	89 (20-100)	88 (20-100)	87 (20-100)	89 (20-100)
ECOG PS, n (%)					
0	179/399 (45)	135/304 (44)	44/95 (46)	63/143 (44)	107/226 (47)
1	171/399 (43)	132/304 (43)	39/95 (41)	65/143 (45)	86/226 (38)
2	49/399 (12)	37/304 (12)	12/95 (13)	15/143 (11)	33/226 (15)
Immunophenotype (%)					
Pro-B	57 (13)	57 (17)		10 (6)**	46 (19)**
Common	256 (59)	256 (77)		107 (68)**	139 (57)**
Pre-B	19 (4)	19 (6)		9 (6)**	9 (4)**
Pre-T	38 (9)		38 (36)	9 (6)**	29 (12)**
Cortical	38 (9)		38 (36)	19 (12)**	18 (7)**
Mature	4 (1)		4 (4)	2 (1)**	2 (1)**
T, nonspecified	1 (0.2)		1 (1)	1 (1)**	0**
ETP††	23 (5)		23 (22)		

ECOG PS, Eastern Cooperative Oncology Group performance status.

*P ≤ .001.

†P = .006.

‡P = .035.

§P = .033.

¶P = .044.

‡P = .005.

#P = .028.

**P = .002.

††All patients with ETP-ALL were assigned to allo-HSCT after FLAG-IDA.

Table 3. Distribution of patients with BCP-ALL according to the genetic subtype (n = 282 patients with available genetic data)

Genetic subgroup	BCP-ALL (n = 282)
<i>BCR::ABL1</i> -like	41 (15%)
<i>KMT2Ar</i>	30 (11%)
Low hypodiploid	22 (8%)
High hyperdiploid	18 (6%)
<i>TCF3::PBX1</i>	14 (5%)
<i>PAX5 P80R</i>	14 (5%)
<i>ZNF384r</i>	10 (3.5%)
<i>IGHr</i>	7 (2.5%)
<i>PAX5alt</i>	6 (2%)
<i>DUX4r/ERGdel</i>	4 (1.5%)
<i>CEBP/ZEB2</i>	4 (1.5%)
<i>iAMP21</i>	4 (1.5%)
<i>CDX2/UBTF</i>	3 (1.5%)
<i>ETV6::RUNX1</i>	2 (0.7%)
<i>MYCr</i>	2 (0.7%)
<i>MEF2Dr</i>	2 (0.7%)
Near haploid	1 (0.3%)
B-other	97 (34%)

CDX2, caudal type homeobox 2; *CEBP*, CCAAT/enhancer-binding proteins; *DUX4r*, double homeobox 4 gene rearranged; *ERGdel*, ETS-related gene deleted; *iAMP21*, intrachromosomal amplification of chromosome 21; *KMT2Ar*, *KMT2A*-lysine methyltransferase 2A gene rearranged; *MEF2Dr*, myocyte enhancer factor 2D gene rearranged; *MYCr*, *MYC* gene rearranged; *PAX5alt*, paired box 5 gene altered; *PAX5 P80R*, paired box 5 gene with P80R mutation; *UBTF*, upstream binding transcription factor gene; *ZEB2*, zinc finger E-box binding homeobox 2 gene.

HyperCVAD was given to 4 of 5 patients with refractory disease (the remaining received alternative therapy), with only 1 response.

Therapy assigned after I-1

By ITT, 157 of 400 (39%) patients with CR after I-1 were assigned to consolidation and maintenance, either due to MRD of <0.01% and no HGR (n = 109) or to MRD of <0.01% and unavailable genetic risk (n = 48). In turn, 243 of 400 (61%) patients were assigned to allo-HSCT either due to MRD of <0.01% but HGR (n = 64), to MRD of ≥0.01% regardless of the genetic risk (n = 127), or to refractoriness to I-1 (n = 52).

Consolidation was given to 18 patients with ETP-ALL, 3 of them relapsed and 1 was on therapy at analysis. Per protocol, allo-HSCT was performed in 14 patients (10 with MRD of <0.01%), of whom 1 experienced disease relapse, 2 died from transplant-related mortality (TRM), and 11 were alive in CR.

CIR and OS by ITT and per protocol

Relapses occurred in 124 of 387 (32%) patients (BM [n = 88], CNS [n = 8], BM and CNS [n = 9], BM and other extramedullary

[n = 11] and other extramedullary-only [n = 8]). Relapses occurred during treatment in 54 patients and off therapy in 41 (after end of chemotherapy [n = 5] and after allo-HSCT [n = 36]) and in 29 after deviation or withdrawal of the protocol. The 3-year CIR was 39% (95% confidence interval [CI], 33-45; Figure 3A). The CIR of patients assigned to chemotherapy (n = 157) and to allo-HSCT (n = 230) were 30% (95% CI, 22-38) and 46% (95% CI, 38-54), respectively (P = .003; Figure 3B). No differences were observed according to age (18-40 years, 41% [32%-49%]; 40-60 years, 37% [29%-45%]) and phenotype (BCP-ALL, 41% [32%-50%]; T-ALL, 48% [35%-60%]; Figure 3C-D). However, significant differences were observed according to the phenotype in both BCP-ALL (Figure 3E) and T-ALL (Figure 3F), with the most immature phenotypes showing the highest CIR. Supplemental Figure 1A, available on the Blood website, shows the CIR observed in the most prevalent BCP-ALL genetic subgroups. Of note, no relapses were observed among patients with the *PAX5 P80R* and *ZNF384r* genetic subtypes, in contrast with the highest CIR shown among patients with *TCF3::PBX1* and *KMT2A* rearrangement. Regarding T-ALL, there were no significant differences in CIR according to the genetic risk, as defined by the protocol (standard genetic risk [SGR], 41% [21%-60%] vs HGR, 48% [27%-66%]; P = .568). For ETP-ALL, the 3-year CIR was 25% (8%-48%; Figure 4B).

White blood cell (WBC) count of >30 × 10⁹/L, HGR, and MRD of ≥0.01% after I-1 emerged as the main prognostic factors for CIR in univariate analysis, whereas WBC count and HGR were associated with increased CIR in multivariable analysis (supplemental Figure 2A-B). When the analysis was restricted to BCP-ALL only, HGR was associated with increased CIR by multivariable analysis (hazard ratio, 1.918 [1.231-2.986]; P = .004).

After a median follow-up of 2.0 years (range, 0.3-5.1), 121 patients had died and 292 remained alive, with a 3-year OS probability of 64% (58%-69%; Figure 5A). Deaths attributable to treatment toxicity (n = 47) occurred in induction (n = 18), during early consolidation/reinduction (n = 6), delayed consolidation (n = 1), and after allo-HSCT (n = 22). The 3-year OS among patients assigned to chemotherapy (n = 157) and to allo-HSCT (n = 243) was 81% (72%-87%) and 54% (46%-62%), respectively (P < .001; Figure 5B). No differences were observed according to age (18-40 years, 67% [59%-74%]; 40-60 years, 60% [52%-68%]) and phenotype (BCP-ALL, 65% [58%-71%]; T-ALL, 60% [46%-72%]; Figure 5C-D). Significant differences were observed in OS according to the phenotype within both BCP-ALL (Figure 5E) and T-ALL (Figure 5F), with the most immature phenotypes (pro-B-cell and pre-T-cell) showing the poorest OS. Supplemental Figure 1B shows OS observed in the most common BCP-ALL genetic subgroups. Of note, all patients with *PAX5 P80R* and *ZNF384r* genetic subtypes survived. The OS probability of patients with *BCR::ABL1*-like subtype was 62% (44%-75%), which was not significantly different from that of the remaining patients with BCP-ALL (supplemental Figure 1C); 41 (88%) of these patients were assigned to allo-HSCT according to poor MRD clearance and/or HGR. OS in T-ALL according to the genetic risk were 72% (50%-86%) for SGR vs 46% (25%-65%) for HGR (P = .150). The 3-year EFS was 49% (43%-54%) and the 3-year RFS was 51% (45%-57%) (supplemental Figure 3A,C). For ETP-ALL, the 3-year OS was 61% (37%-79%; Figure 4A).

Table 4. Main clinical, biological, and outcome characteristics of the most frequent genetic subtypes (including 10 or more patients) of BCP-ALL and T-ALL

Genetic subgroup	n (%)	Age, y median (range)	Sex M/F	WBC count, ×10 ⁹ /L median (range)	CNS infiltration n/N(%)	CR after I-1 (%)	MRD of <0.01% after I-1 (%)	3-y CIR (95% CI)	3-y OS (95% CI)
BCR::ABL1-like	41 (15)	40 (18-60)	26/15	32 (0.8-561)*	10/39 (26)†	75	20	37% (17-58)	62% (44-75)
KMT2Ar	30 (11)	44 (20-60)	10/20	140.8 (2-20)‡	3/30 (10)	100	60	63% (39-80)	26% (8-47)
Low hypodiploid	22 (8)	51 (29-60)§	13/9	2.9 (0.9-36.7)	1/21 (5)	68	43	35% (14-57)	48% (25-68)
High hyperdiploid	18 (6)	36 (18-55)¶	11/7	3.3 (0.7-19.3)#	2/15 (13)	100	83	12% (2-32)	80% (49-93)
TCF3::PBX1	14 (5)	41 (18-52)	5/9	13.2 (5.4-79.1)	2/12 (17)	100	86	57% (28-78)	76% (43-92)
PAX5 P80R	14 (5)	38 (18-60)	8/5	10.2 (0.8-56)	0	100	100	0%	100%
ZNF384r	10 (3.5)	34 (18-48)	5/5	7.3 (1.3-20)	1/9 (11)	90	40	0%	100%
SGR non-ETP T-ALL	31/65 (48)	29 (18-60)	20/11	36.4 (4.8-551.4)	3/31 (10)	90	71	41% (21-60)	72% (50-86)
HGR non-ETP T-ALL	34/65 (52)	35 (18-59)	24/10	33.1 (1.2-740)	7/30 (23)	74	72	48% (27-66)	46% (25-65)

All patients with KMT2Ar and low hypodiploidy had HGR by trial definition. HGR for the remaining subtypes were as follows: BCR::ABL1-like (16/41 [39%]), high hyperdiploidy (0/16; 2 not available), TCF3::PBX1 (1/13 [7.7%]; 1 not available), PAX5 P80R (2/12 [16.7%]; 2 not available), and ZNF384r (1/10 [10%]). In patients with BCR::ABL1-like ALL, those with HGR had significantly inferior OS and higher CIR than those with SGR: 36% (12%-61%) vs 78% (55%-90%), $P = .020$, for 2-year OS; and 48% (19%-73%) vs 14% (2%-39%), $P = .030$, for 2-year CIR.

CR, complete remission; F, female; KMT2Ar, KMT2A-lysine methyltransferase 2A gene rearranged; M, male; PAX5 P80R, paired box 5 gene with P80R mutation.

* $P = .010$.

† $P = .014$.

‡ $P < .001$.

§ $P = .001$.

|| $P = .041$.

¶ $P = .038$.

$P = .015$.

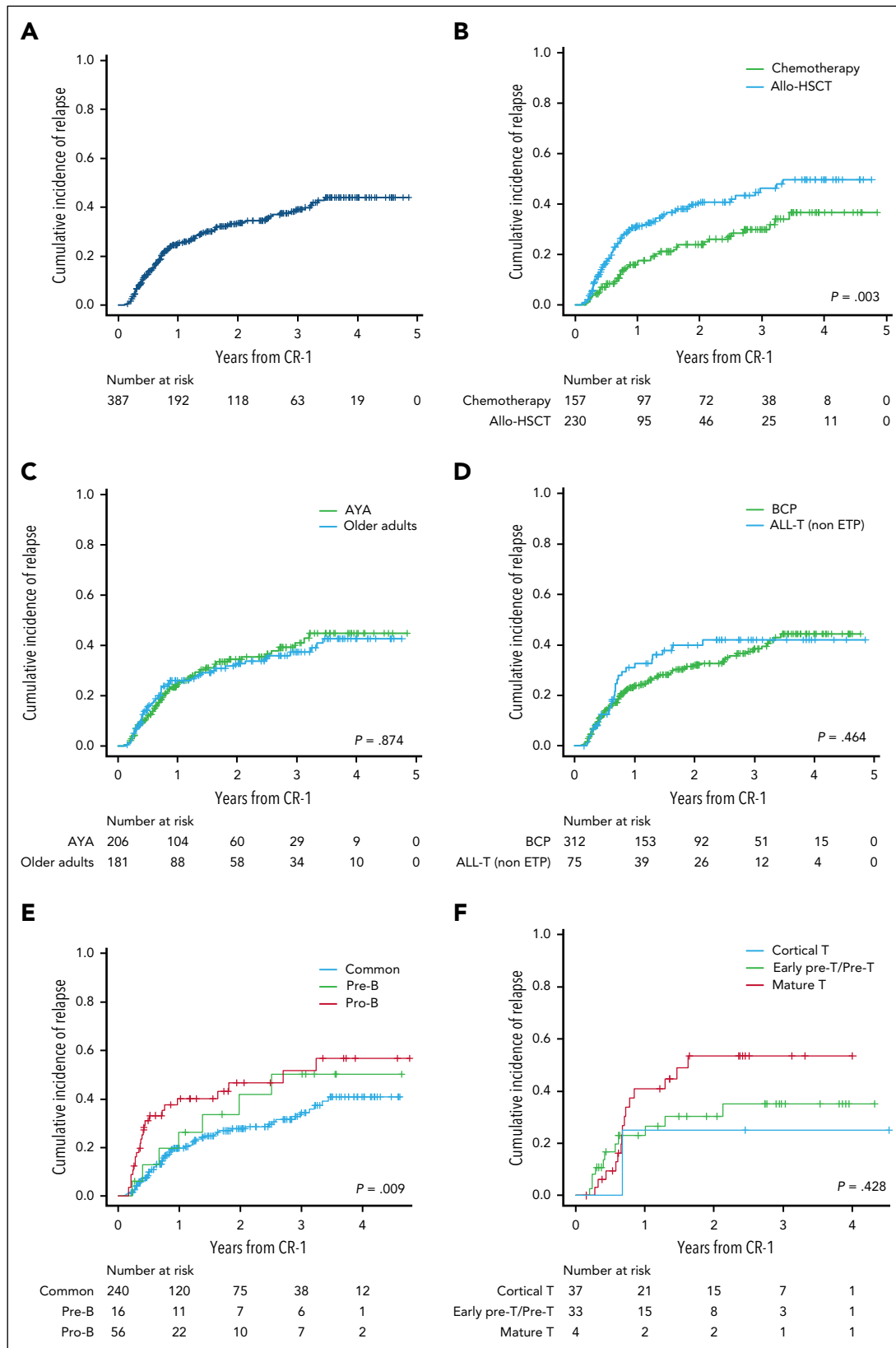


Figure 3. CIR by ITT in patients with BCP-ALL and T-ALL, excluding ETP. (A) The CIR in the whole series; (B) the CIR according to the postremission assigned schedules, being significantly lower for patients assigned to chemotherapy (vs allo-HSCT). The CIR according to age (C) and phenotype (B vs T) (D). (E) The CIR according to the immune phenotype subtypes within BCP-ALL; (F) the CIR according to the T-ALL subtypes excluding ETP-ALL. Patients with common ALL and those with cortical and mature T-ALL showed lower CIR than those with pro-B-ALL and pre-T-ALL, respectively. AYA, adolescents and young adults; CR-1, first CR.

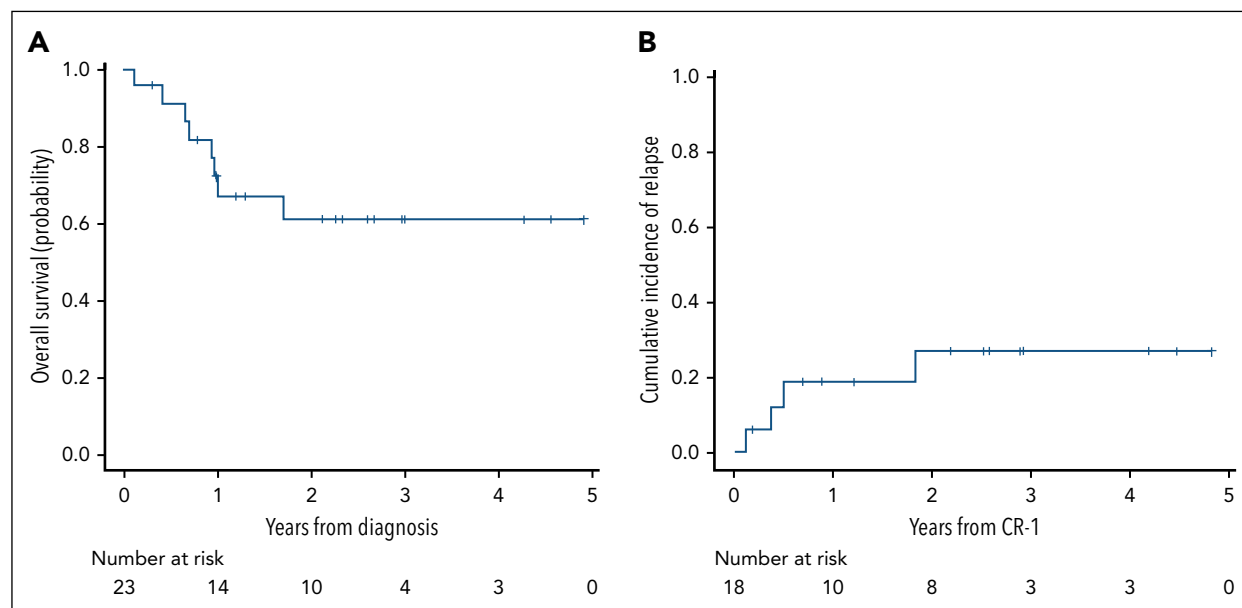


Figure 4. OS and CIR in 23 patients with ETP-ALL. (A) The 3-year OS probability was 61% (95% CI, 37-79); and (B) 3-year CIR was 25% (95% CI, 8-48). CR-1, first CR.

WBC count of $>30 \times 10^9/L$, HGR, and MRD of $\geq 0.01\%$ after I-1 emerged as the main prognostic factors for OS in univariate analysis, whereas HGR was the only poor prognostic factor in the multivariable analysis (supplemental Figure 2C-D). When the analysis was restricted to BCP-ALL, only HGR was associated with poor OS by multivariable analysis (hazard ratio, 2.182 [1.272-3.742]; $P = .005$).

The 3-year OS rate of 327 patients treated per protocol was 66% (59%-72%) and the CIR was 37% (30%-43%); these outcomes were 83% (74%-90%) and 28% (20%-37%), respectively, for patients in the chemotherapy arm, and 57% (47%-65%) and 44% (35%-54%), respectively, in the allo-HSCT arm ($P < .001$ for OS, and $P = .018$ for CIR; supplemental Figure 4A-B). In patients who received allo-HSCT ($n = 178$) the rate of MRD of $<0.01\%$ before transplantation was 92% (154/167). The 3-year CIR and TRM were 40% (30%-50%) and 14% (9%-20%), respectively (supplemental Figure 5).

Outcomes of patients according to genetic risk and MRD response after I-1

Figure 6A shows the OS probability according to combination of the baseline genetic risk and the MRD levels after I-1. Of note, patients with MRD of $<0.01\%$ without HGR ($n = 109$) had 3-year OS probability of 81% (70%-89%) and a CIR of 32% (22%-42%), whereas OS among patients with MRD of $<0.01\%$ and HGR ($n = 64$) was 50% (34%-63%; $P < .001$), a percentage similar to that of patients with MRD of $\geq 0.01\%$. Patients with SGR, EOI-1 MRD of $<0.01\%$, and MRD of $<0.001\%$ at the end of early consolidation/reinduction ($n = 70$) showed 3-year OS of 89% (78%-95%; Figure 6B) and a CIR of 19% (10%-31%).

Toxicity

Grade 3 to 5 toxicities in I-1, I-2, and early consolidation/reinduction are summarized in supplemental Table 1. Hematologic toxicity and infections were the most frequent. Infection by severe acute respiratory syndrome coronavirus 2 was detected

in 66 patients, being fatal in 2. The most frequent toxicity by PEG-ASP was hepatic, especially during induction (22%; supplemental Table 2). Thrombosis of any grade was registered in 13% of patients during induction. Silent inactivation was detected in 9 of 204 (4%) monitored patients. Dose modifications of PEG-ASP were registered in 13% of patients during induction and in 39% of patients in consolidation/reinduction.

Comparison with the outcomes of the ALL-HR-11 trial⁴

Patients from ALL-HR-11 and ALL-2019 trials were comparable for the main clinical and biologic ALL characteristics (data not shown). There was a significant improvement in OS in the ALL19 trial compared with the ALL-HR-11 (3-year OS: 64% [58%-69%] vs 57% [50%-62%]; $P = .038$) despite showing similar CIR (38% [33%-44%] vs 41% [35%-47%]; $P = .837$; Figure 7). Death rate in induction (5% vs 6%) and consolidation (5% vs 4%), and CR rate (94% vs 91%), were similar. In contrast, the CI of TRM was lower in the ALL-2019 trial (14% vs 24%) and OS from relapse was slightly better in the ALL19 trial (26% vs 21%, $P = .469$). No differences were observed on comparison of EFS and RFS between the 2 protocols (3-year EFS: 49% [43%-54%] for ALL-2019 vs 43% [36%-49%] for ALL-HR-11 trial, $P = .249$; 3-year RFS: 52% [46%-57%] vs 47% [40%-53%], $P = .513$; supplemental Figure 4B,D).

Discussion

This prospective multicenter study shows that the combination of genetics and MRD allows accurate identification of adult patients with Ph⁺ ALL as candidates for consolidation with allo-HSCT or chemotherapy, and allows identification of a subgroup of patients without HGR and undetected/very low MRD levels, who show excellent prognosis without transplantation.

Baseline cytomic features and MRD kinetics are strong prognostic factors in ALL, but whether early MRD clearance can

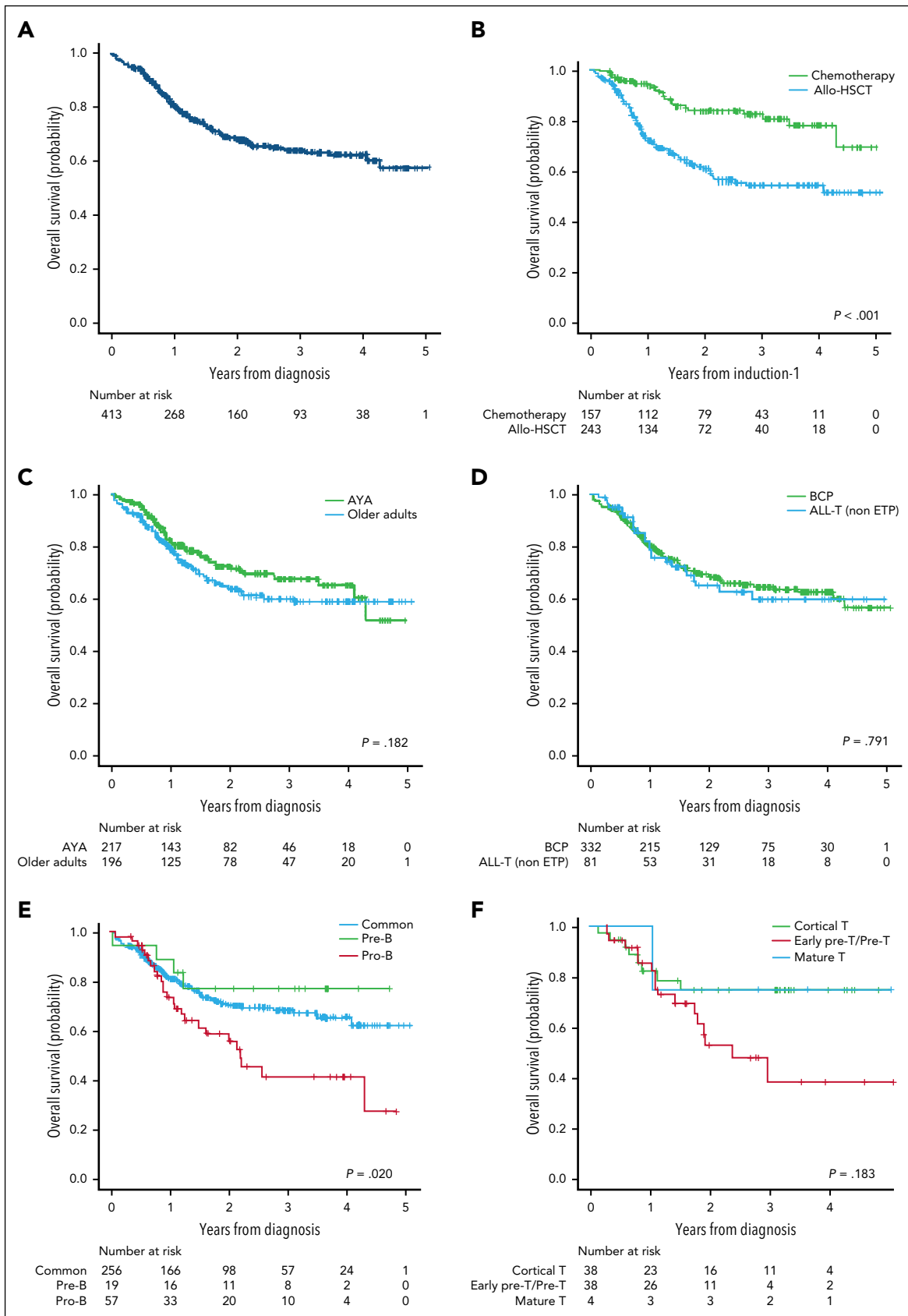


Figure 5. OS by ITT in patients with BCP-ALL and T-ALL, excluding ETP-ALL. (A) The OS in the whole series; (B) the OS according to the postremission assigned schedules, being significantly better for patients assigned to chemotherapy (vs allo-HSCT). The OS according to age (C) and phenotype (B vs T) (D). (E) The OS according to the immune phenotype subtypes within BCP-ALL; (F) the OS according to the T-ALL subtypes excluding ETP-ALL. The poorest OS was observed in the immature subtypes of B and T lineage (pro-B and early pre-T/pre-T), respectively. AYA, adolescents and young adults.

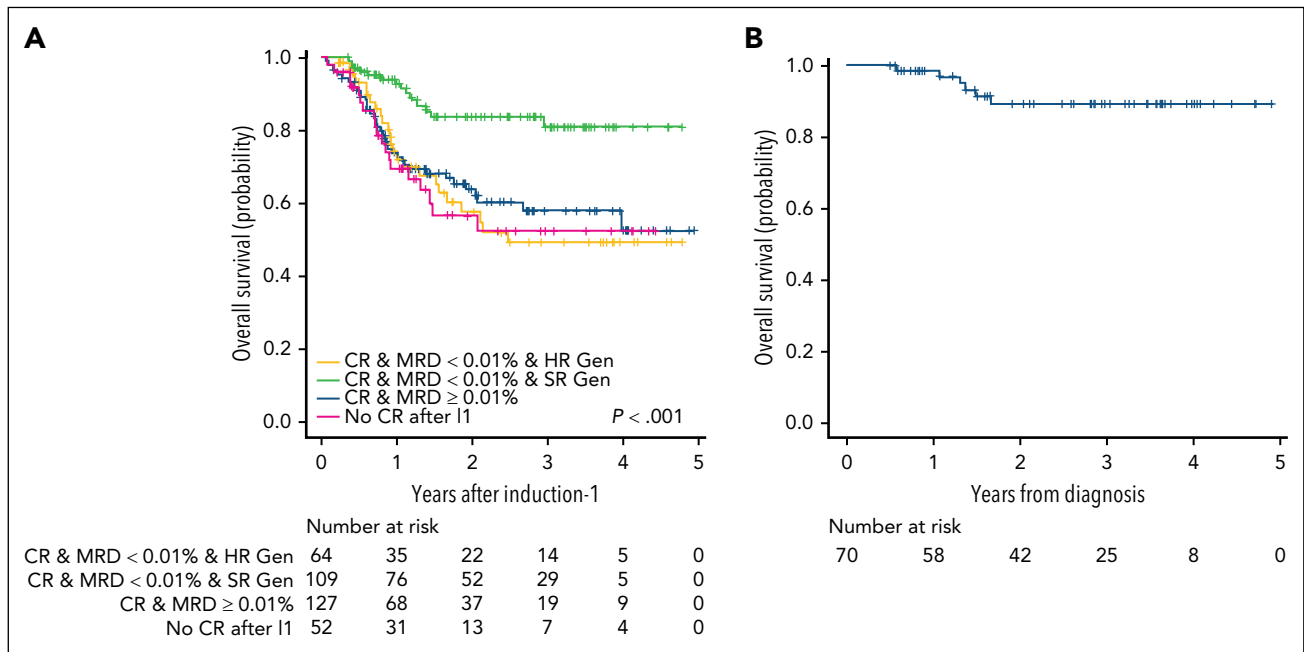


Figure 6. OS probabilities according to the combination of the genetic risk and the MRD level at the end of I-1 in patients with BCP-ALL and T-ALL excluding ETP-ALL. Panel A compares the OS of 4 groups of patients: (1) CR after I-1, MRD < 0.01%, and HGR; (2) CR after I-1, MRD < 0.01%, and SGR; (3) CR after I-1, and MRD ≥ 0.01%; and (4) no CR after I-1. (B) The OS of patients with CR after I-1, SGR, and MRD of < 0.01% at the end of I-1 who had MRD level of < 0.001% at the end of early consolidation and reinduction. Significant differences were observed comparing groups 1 (that accounts for 31% of the patients) and 2 (accounting for 18% of patients): 3-year OS probability of 81% (95% CI, 70-89) vs 50% (95% CI, 34-63), respectively; $P < .001$. Gen, genetics.

offset the poor prognosis of high-risk genetic subtypes remains uncertain. The prognostic value of early MRD kinetics combined with genetic risk has scarcely been investigated in newly diagnosed Ph⁻ ALL. In this study, the MRD cutoff for early MRD response and the HGR followed established consensus guidelines.¹⁹⁻²¹ Notably, BCR::ABL1-like ALL was not classified as HGR, although 80% of such patients underwent allo-HSCT due to poor early MRD clearance and/or HGR-associated features.

Outcomes in this trial were comparable with those of contemporary studies not incorporating frontline immunotherapy.^{22,23} The 83% rate of MRD of < 0.01% with the FLAG-IdA regimen achieved in MRD-positive patients after I-1 is of note. These patients did not receive blinatumomab, which was not approved for MRD-positive patients in Spain during the study period. High CR rates were obtained in patients with I-1–refractory disease treated with FLAG-Ida or InO (82% and 92%,

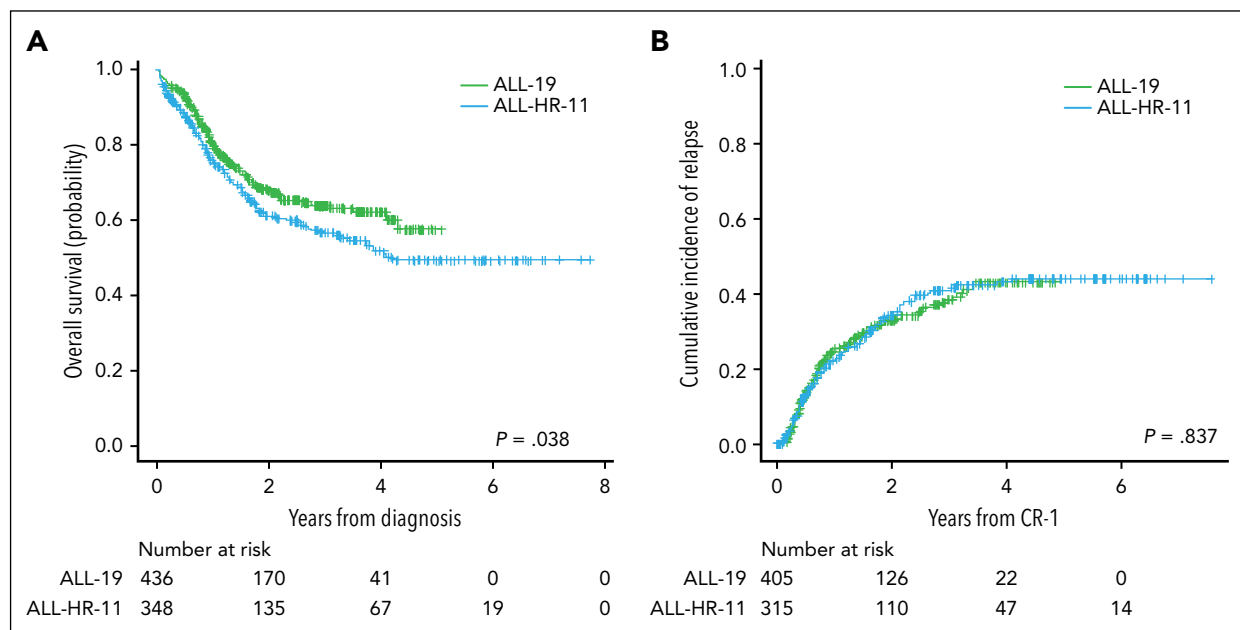


Figure 7. Comparison of the OS and CIR between the current trial (ALL-2019) and the previous PETHEMA trial (ALL-HR-11). There was a significant improvement in OS in the ALL19 trial (A), despite similar CIR (B). CR-1, first CR.

respectively), confirming the efficacy of both regimens in this setting.²⁴

The 3-year CIR for patients assigned to chemotherapy by ITT was 30%. Despite being lower than that of patients who received allo-HSCT, this rate is still high. In patients who underwent allo-HSCT, the 14% TRM was acceptable for a multicenter study, although the high relapse rates indicate room for improvement. Immunotherapy combined with chemotherapy improves the frequency and depth of MRD response, and reduces the need for allo-HSCT, as shown in recent trials.^{8,25,26}

In patients without HGR, early CR achievement and early MRD response allowed the identification of a subgroup of patients with 3-year OS probability of 81%, compared with 50% for patients with similar early MRD response but HGR, suggesting an independent impact of genetic characteristics on outcome.²⁷ This was confirmed by the multivariable study, which identified HGR as the only variable associated with poor OS. Survival was even better (89%) among patients with MRD of <0.001% at end of early consolidation/reinduction. However, despite this promising outcome, a 3-year CIR of 19% was observed, reinforcing the need for improved frontline therapies also in this favorable group.

The FLAG-Ida schedule was used as I-1 for patients with ETP-ALL, given their poor outcome shown in previous trials from our group^{4,28,29} and based on the concept of its very immature origin and the presence, per definition, of some myeloid markers (eg, CD13 or CD33).^{30,31} All patients with ETP-ALL showed HGR.³⁰ The outcome was poorer than that of cortical or mature T-ALL and was similar to that observed in pre-T-ALL, supporting the need for improved therapies for patients with immature T-ALL subtypes.³²

A modest but statistically significant improvement in OS was observed compared with the previous ALL-HR-11 trial,⁴ despite showing similar CIR. Several factors can explain this improvement, of which a slightly higher CR rate, lower TRM, and better results of rescue therapy are the most important. Despite this modest improvement, some limitations should be pointed out. First, centralized genetic risk was not available in 18% of patients, mainly due to the poor quality of the samples or to problems of sending samples for centralized analyses during the COVID-19 pandemic. These patients were assigned to chemotherapy or allo-HSCT based on the EOI-1 MRD level. Second, there were protocol deviations, especially in patients allocated to allo-HSCT, similarly to other studies.²³⁻²⁶ However, the per protocol analysis, excluding all deviations, showed almost identical results. Third, the proportion of patients in the B-other group was higher than that observed in recent studies,³³ suggesting the need for improved genetic characterization using more sensitive techniques, such as optical genome mapping and RNA sequencing. Fourth, the *BCR::ABL1*-like ALL subtype was not considered as HGR in our study, although 88% of patients were allocated to allo-HSCT by the combination of HGR and poor early MRD clearance. Their high transplant rate could explain the similar prognosis compared with that of the remaining patients with BCP-ALL.³⁴ Fifth, the MRD level was not predictive of outcome in some genetic subtypes, such as *KMT2A*-rearranged ALL, suggesting that the genetic driver event might target an earlier precursor

cell that might go undetected by FCM despite showing the *KMT2A* gene rearrangements.^{33,35} Sixth, because treatment assignment by MRD criteria was done at EOI-1, the possible effect of MRD at later time points (eg, week 10-16) could not be assessed. Finally, the genetic risk for T-ALL used in this trial was not useful for prognostic discrimination. Improved risk classifications, not available at the trial design, have proven useful for prognostic assessment in T-ALL.^{32,36,37}

This prospective multicentric study showed that the combination of baseline genetics and early MRD clearance allows an accurate postremission assignment of treatment for patients with Ph⁻ ALL, and confirms the independent prognostic significance of genetics, especially in BCP-ALL. Because no further improvements are expected with the use of standard chemotherapy with or without allo-HSCT, the role of MRD and genetics should be reassessed in current and future studies incorporating immunotherapy and other targeted therapies in the frontline treatment of patients with Ph⁻ ALL.³⁸

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Authorship

Contribution: J.-M.R. designed and led the study, analyzed the data, and wrote the manuscript; A.T. was coprincipal investigator, designed and conducted the study, and edited the manuscript; M.M. analyzed the data, and edited the manuscript; S.B., J.R., E.G., A.H.-S., R.S., E.S., M.H., and J.C. performed the centralized biologic studies (immunophenotype, single-nucleotide polymorphism arrays, next-generation sequence panels, and RNA-sequencing panel for Philadelphia chromosome-like acute lymphoblastic leukemia); I.G., J.-M.H.-R., and

A.O. performed the centralized biologic studies and coordinated the biological group; E.Z. performed the centralized analysis of asparaginase activity; J.G.-C., N.M.-C., P.M., P.B., A.N., M.S., C.M., C.B., M.-P.Q.d.L., M.C., A.V., P.L.d.U., M.T., B.-J.G.-G., I. García-Cadenas, C.R.-M., P.M.-S., A.B., I. Gómez-Centurión, L.H., J.S., J.-A.R., L.S., V.R.-G., A.d.L., L.G.-P., M.S.-R., M.-S.C., L.L., D.B., F.V.-I., J.-L.A., M.-L.A., R.G.-B., A.C., J.B., M.-J.C., and E.F. managed the study; and all the authors revised and approved the manuscript.

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